

Smart Kitchen Intelligence

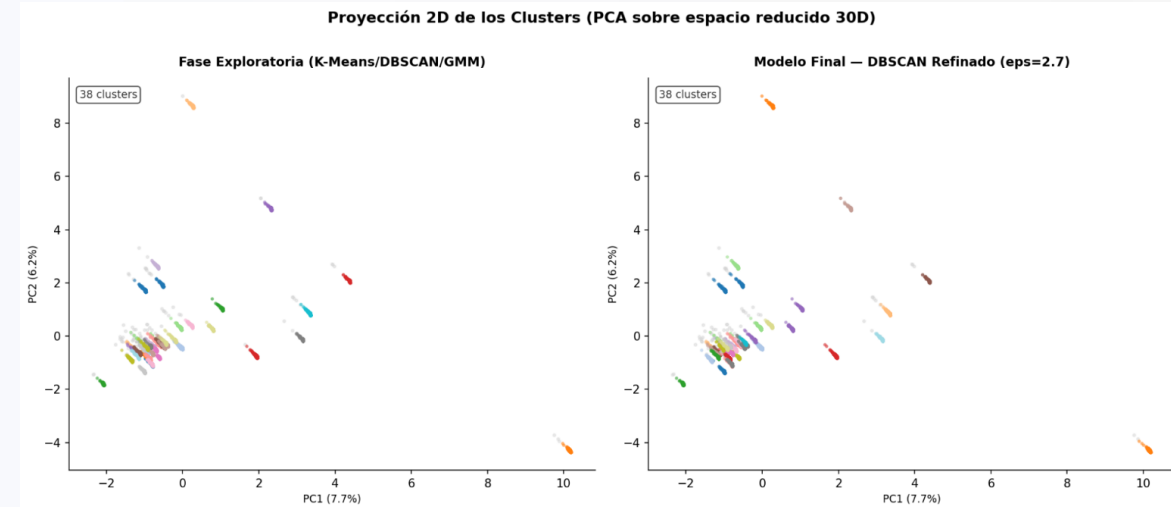
Final Integration, Orchestration, and Product
Deployment

Authors: José Guillermo Melgar Puertas & Gabriel Alonso Reyna Alvarado
Course: Big Data · Computer Science Department (UPC)
Professor: Carlos Adrian Alarcon Delgado · July 2026

🕒 Operational Background (Milestones 1 to 4)

Data Pipeline Foundations

- ▶ Ingestion and Quality (M1 & M2): Ingested 25,445 parameterized stochastic events (Instacart + USDA API) into a unified Star Schema mapped to inventory_v1.csv.
- ▶ Feature Engineering (M2 & M3): Discrete temporal mapping using flat hour_of_day, PCA dimensional reduction to 30 components preserving 90.01% of total variance.
- ▶ Lifestyles Segmentations (M3): Discovered 38 stable demographic cohorts via DBSCAN (), isolating anomalies to the afternoon hours (14.0h) due to categorical food dispersion.
- ▶ Baseline Algorithms (M4): Semantic matching (TF-IDF), collaborative replenishment (iALS with regularization), and physical expiry tracking.



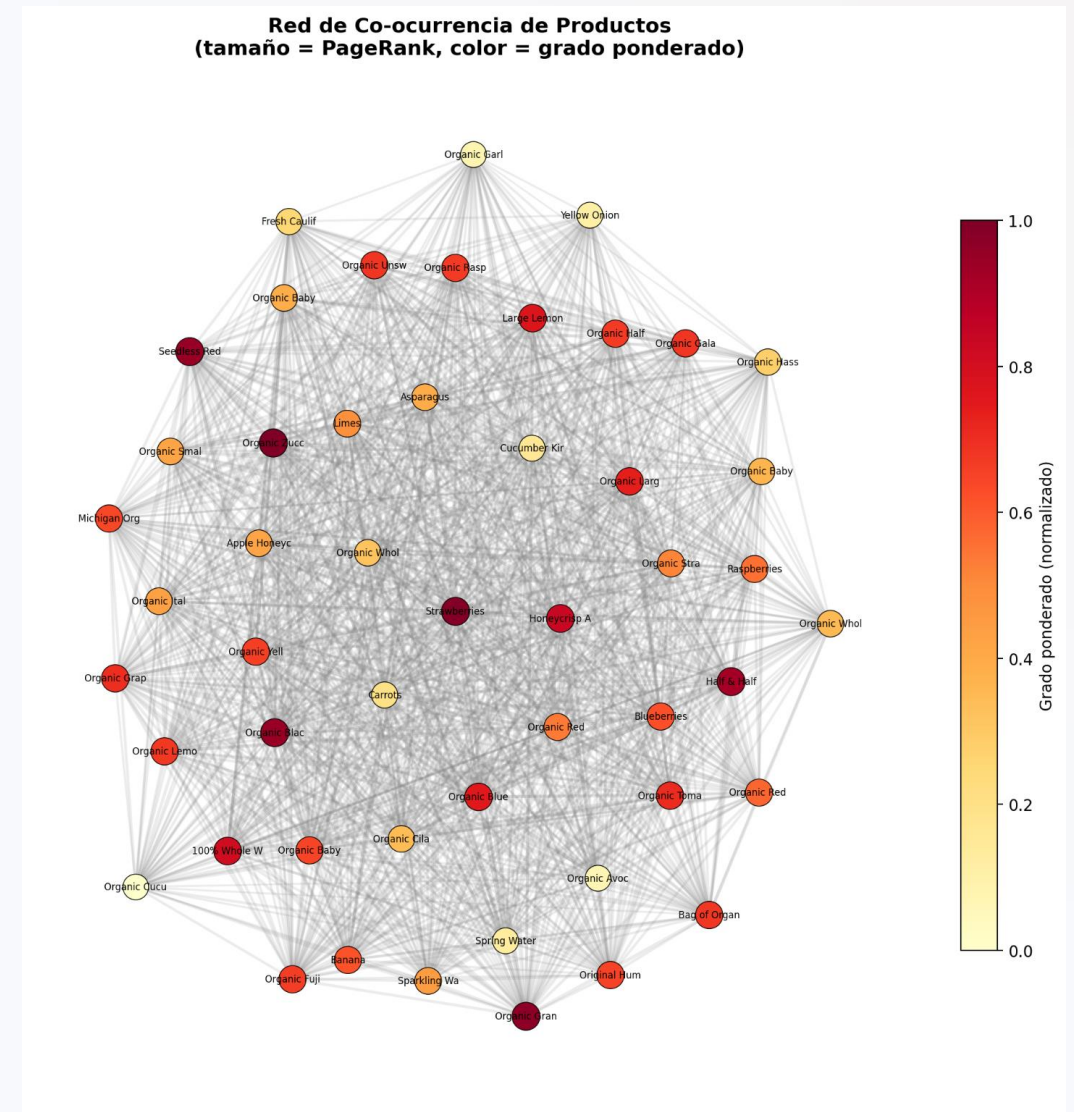
Network Layer: Co-Occurrence & PageRank

Mapping Structural Kitchen Co-Consumption

- ▶ Undirected Weighted Graph: Modelled the kitchen catalog containing 50 item nodes and 1,225 potential interaction edges (dense network).
- ▶ Adjacency Projection Matrix: Quantified the co-occurrence through a cross-multiplication of the binary replenishment session matrix :

$$W = R_{\text{bin}}^T R_{\text{bin}} \quad \text{with } W_{ii} = 0$$

- ▶ Markov PageRank Spectral Centrality: Solved with damping factor $d=0.85$. Showed extremely low variance ($\sigma \approx 0.00094$). Percibles (fruits/veggies) act as pivotal network bridges.

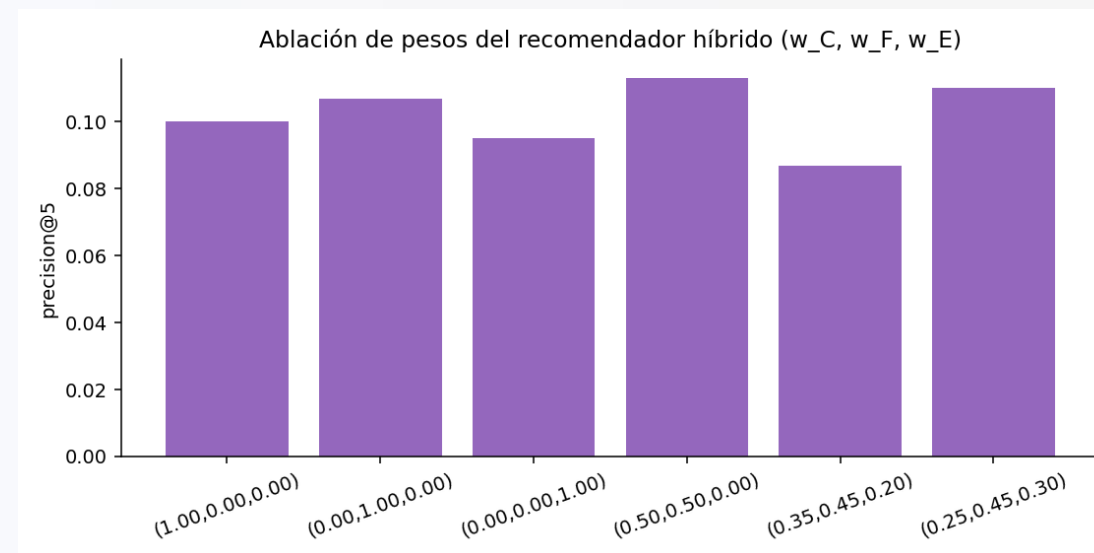


⚙️ Integrated Hybrid Recommendation Engine

Linear Multi-Key Convex Ensemble

$$S_{\text{hybrid}} = w_C S_{\text{content}} + w_F S_{\text{cf}} + w_E S_{\text{expiry}} + w_G S_{\text{pagerank}}$$

- ▶ Production Weights (hybrid_meta.json): Calibrated offline in the 4-simplex via cross-validation: (0.35 , 0.45 , 0.20 , 0.00)
- ▶ PageRank Decoupling Design: Forcing a linear weight saturated recommendations with hyper-popular global items, destroying household personalization.
- ▶ Architectural Solution: PageRank operates independently as the UI engine powering the "Pantry Essentials" panel.



Evaluation Protocol & Algorithm Benchmark

Algorithmic Serving Model	Precision@5	MAP@5	Catalog Coverage
Popularity Baseline	0.0508	0.0539	22.00%
PageRank Centrality (Structural)	0.0541	0.0594	26.00%
Content-Based TF-IDF	0.0450	0.0511	84.00%
Collaborative Filtering (iALS)	0.0467	0.0549	100.00%
SKI Hybrid Model (Production)	0.0494	0.0574	100.00%
Experimental v2_graph Hybrid	0.0479	0.0560	100.00%

The Defense Argument: Although the static PageRank baseline yields slightly higher absolute MAP@5 (0.0594) by recommending the same popular ingredients, it suffers from a critically low 26% coverage and lacks temporal/expiry alerts. The SKI Hybrid Model achieves optimal personalized relevance (MAP@5 = 0.0574) while maintaining 100% catalog coverage and anti-waste urgency tracking.

⚙️ Automation, Quality Assurance & Reproducibility

Industrializing the Backend Core

- ▶ Orchestrated Scripting: Consolidated 19 manual fragmented pipeline files into a single end-to-end execution flow: run_pipeline.py.
- ▶ One-Line Reproducibility: Guarantees clean structural rebuilds of all physical matrices, vectors, and weights inside data/processed/.
- ▶ Rigorous Quality Suite: Implemented 35 unit and regression tests in pytest, actively validating matrix shapes, ranges, and null counts.

```
(venv) PS C:\Users\Usuario\Documents\GitHub\Smart-Kitchen-Intelligence> python run_pipeline.py --skip-ingestion
[hit01] simulation.py - Simulación de movimientos de inventario (90 días)
=====
🚀 Iniciando simulación masiva: 10 hogares durante 90 días...
🏠 Simulando Hogar ID: 0...
🏠 Simulando Hogar ID: 1...
🏠 Simulando Hogar ID: 2...
🏠 Simulando Hogar ID: 3...
🏠 Simulando Hogar ID: 4...
🏠 Simulando Hogar ID: 5...
🏠 Simulando Hogar ID: 6...
🏠 Simulando Hogar ID: 7...
🏠 Simulando Hogar ID: 8...
🏠 Simulando Hogar ID: 9...

✅ ¡Éxito! Se generaron 25929 eventos transaccionales.
📦 Volumen final: 3.89 MB en disco.
Classification
Purchase      11618
Consumption   8773
Forced_Waste  3933
Waste         1605
Name: count, dtype: int64
[ok] simulation.py completado en 4.3s
```

pytest.

Serving, Telemetry & User Interface

Model Deployment & Monitoring

- ▶ Hybrid Serving Strategy: Weekly high-performance batch computing for iALS factorizations and graphs; Near-Real-Time session projection on client.
- ▶ Data Drift Tracking: Active telemetry in `monitoring_plan.md`, using the Kullback-Leibler distance to detect shifts in temporal shopping habits.
- ▶ User-Centered Interface (`demo_app.py`): Clean translated dashboards with no raw machine learning jargon. Displays an interactive **RdYIGn Treemap** of the kitchen.



Final Defense Conclusions

Quantitative Sustainability

Empirically proved that visual inventory analytics directly reduces household waste. Simulated loss metric (17.5%) matches almost perfectly with the UN World Food Waste Index proxy (17%).

Engineering Maturity

The product demonstrates software engineering excellence by seamlessly orchestrating API ingestion, PCA feature extraction, DBSCAN density profiling, iALS factorization, and graph networks.

Business & Product Impact

Translating heavy, invisible backend mathematical matrix operations into beautiful, simple visual alerts (Treemaps) successfully converts raw database files into actionable and sustainable decisions.

Defensible Framework

The structured model benchmarker (MAP@5 vs. Coverage) provides clear evidence of analytical rigor, justifying our architectural decisions to the academic jury.